Quiz-1

Next week

- Topics covered till this week
- details in google classroom.

Matrix Multiplication

$$\begin{bmatrix} | & | & | \\ c_1 & c_2 & c_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \rightarrow a\begin{bmatrix} | \\ c_1 \\ | \end{bmatrix} + b\begin{bmatrix} | \\ c_2 \\ | \end{bmatrix} + c\begin{bmatrix} | \\ c_3 \\ | \end{bmatrix}$$

column

matrix vector col 1 scaled by a.

$$\begin{bmatrix} | & | & | \\ c_1 & c_2 & c_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} - & r_1 & - \\ - & r_2 & - \\ - & r_3 & - \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ -4 & 7 \end{bmatrix} \begin{bmatrix} 2 & -11 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ -4 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$11 \quad \begin{bmatrix} -11 \\ 7 \end{bmatrix}$$

4 diff ways to multiply. (Next class)

multiply left side → manipulate rows

" right " → " cols.

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & -11 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 4 & -22 \\ 4 & 7 \end{bmatrix}$$

left

$$x + 2y + z = 2$$
$$3x + 8y + z = 12$$
$$4y + z = 2$$

Augmented Matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array}\right]$$

Find 3 pivots, each for 1 row.

all gotta be 0.

pivots → Numbers along diagonal.

Elimination Rules:

1) Find pivots
2) pivots ≠ 0
3) Everything under the pivot is 0

$Ax = b$

$x = A^{-1}b$

row 2 = row 2 − 3 × row 1

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 4 & 1 & 2 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array}\right]$$

U

row 3 = row 3 − 2 × row 2

$A \rightarrow U$ (upper triangle)

programmers tries to optimize how fast to go from A to U.

2 Failure → recovery ✓

↳ " ✗

2 changes

1) swap

2) Elimination.

if swap

$$\begin{bmatrix} 1 & 2 & 1 & : & 2 \\ 0 & 0 & -2 & : & 6 \\ 0 & 4 & 1 & : & 2 \end{bmatrix}$$

✗

$$\begin{bmatrix} 1 & 2 & 1 & : & 2 \\ 0 & -4 & -2 & : & 6 \\ 0 & 4 & 0 & : & 2 \end{bmatrix}$$

we shifted after fixing 1st pivot. for the 3rd 0 pivot we can't swap as my upper pivot changes.

upper triangle Matrix can have 0 above pivot.

$5z = -10$

$\therefore z = -2$

$2y - 2z = 6 \quad [z = -2]$

$y = 1$

$x + 2y + z = 2$

$x + 2 - 2 = 2$

$x = 2$

Must work with the upper row (where the pivot is) to get a zero below it. No other row.

$$\begin{bmatrix} \boxed{1} & 2 & 1 & : & 2 \\ 3 & 8 & 1 & : & 12 \end{bmatrix}$$

to go from here to

here, inverse operation

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 & : & 2 \\ 3 & 8 & 1 & : & 12 \\ 0 & 4 & 1 & : & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & : & 2 \\ 0 & 2 & -2 & : & 6 \\ 0 & 4 & 1 & : & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 & : & 2 \\ 3 & 8 & 1 & : & 12 \\ 0 & 4 & 1 & : & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & : & 2 \\ 0 & 2 & -2 & : & 6 \\ 0 & 4 & 1 & : & 2 \end{bmatrix}$$

Elimination Matrix

E_{21}.

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 & : & 2 \\ 0 & 2 & -2 & : & 6 \\ 0 & 4 & 1 & : & 2 \end{bmatrix}$$

$E_{3,2}$

$$E_{32}(E_{21}A) = U$$

$$\Rightarrow (E_{32}E_{21})A = U$$

$$EA = U$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

E_{21}^{-1}

Gilbert ch 2

Anton " 1